Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9648/03

Paper 3 Applications Paper and SPA Planning Task

21 September 2015

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A:
Consists of 4 Structured Questions
Answer **all** questions.
Write your answers in the **space provided** on the question paper.

Section B:
Consists of 1 SPA Planning Task
Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Section C:
Consists of 1 Free-Response Question.
Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Sections A, B and C are to be submitted **separately**.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [ ].

For Examiner's Use	
Section A [40]	
1	**/ 15**
2	**/ 5**
3	**/ 10**
4	**/ 10**
Section B [12]	
Section C [20]	
Total [72]	

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For Examiner's use

Section A: Structured Questions (40 marks)
Answer **all** questions in this section.

Question 1

(a) **Fig. 1.1** below shows a schematic diagram of targeted gene replacement in a mammalian cell involving the gene *Ig M*. Targeted *Ig M* gene replacement is a method of replacing *Ig M* mutated allele (Fig. 1.1a) in the genome in a cell with wildtype allele (Fig. 1.1b) via crossing over, giving rise to the recombinant DNA (Fig. 1.1c).

The wildtype allele is carried by a linearised plasmid (Fig. 1.1b) and introduced into the cell. As shown in **Fig. 1.1b**, the linearised plasmid contains an antibiotic (neomycin) resistance gene (*Neo gene)*, wildtype alllele with the exons represented by the black boxes, as well as several restrictions sites in between the exons. In this technique, the enhancer of the *Neo* gene on the linearised plasmid was removed prior to the therapy.

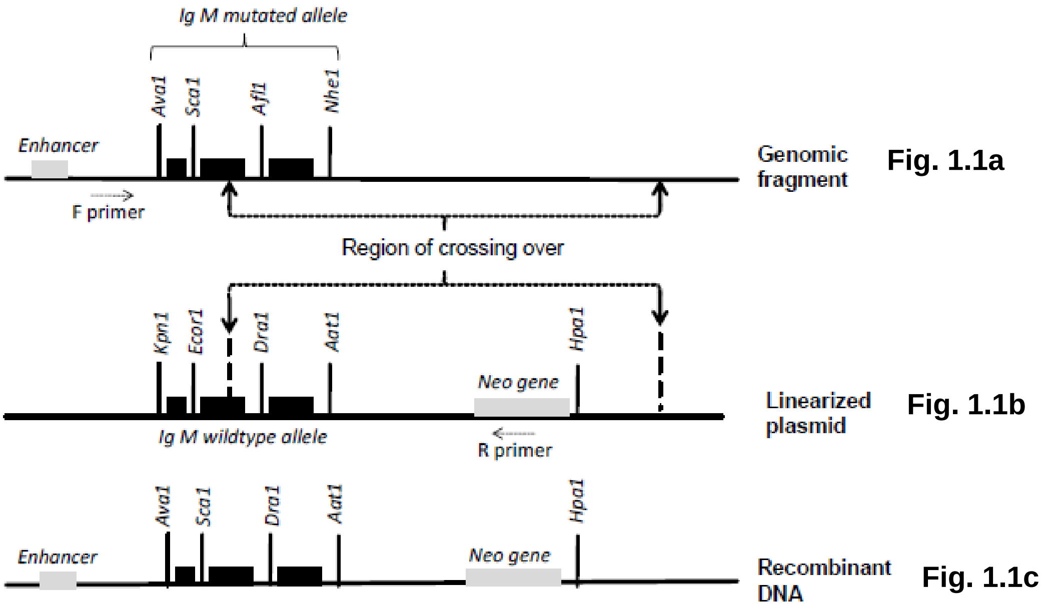

Fig. 1.1

Legend	
Exon	■
Restriction sites on mutated allele	*Ava1, Sca1, Afl1, Nhe1*
Restriction sites on wildtype allele	*Kpn1, Ecor1, Dra1, Aat1*
Naturally occurring restriction site on linearised plasmid	*Hpa1*

(a) **(i)** Explain the significance of restriction enzyme in bacteria. [1]

....

....

(ii) Although the mutated and wildtype alleles have homologous regions, different restriction sites are present on them. Explain the presence of these different restriction sites. [2]

(iii) With reference to **Fig. 1.1**, explain how the Neo gene allows the transformed cells with recombinant DNA to be distinguished from the transformed cells which did not undergo recombination in their DNA. [4]

(iv) With reference to **Fig. 1.1**, briefly describe another method how restriction enzyme *Hpa1* and a probe that is complementary to the enhancer sequence can be used to verify the identity of cells with recombinant DNA. [4]

(b) In the gene replacement therapy shown in **Fig. 1.1**, the site of DNA breakage occurred between restriction sites Sca1 and Afl1 as shown in **Fig. 1.1a**.

In another case of therapy, the site of DNA breakage along the mutated allele was unknown. To determine the exact site of crossing over between the genomic DNA and the linearised plasmid, RFLP analysis was carried out on the recombinant DNA.

The recombinant DNA was isolated from the transformed cell and amplified using PCR via the forward and reverse primers as shown in **Fig. 1.1a** and **Fig. 1.1b**. The amplified DNA products were separated into eight tubes and digested using eight restriction enzymes respectively: Ava1, Sca1, Afl1, Nhe1, Kpn1, Ecor1, Dra1, Aat1. The products of digestions were separated using gel electrophoresis and the results are shown in **Fig. 1.2**.

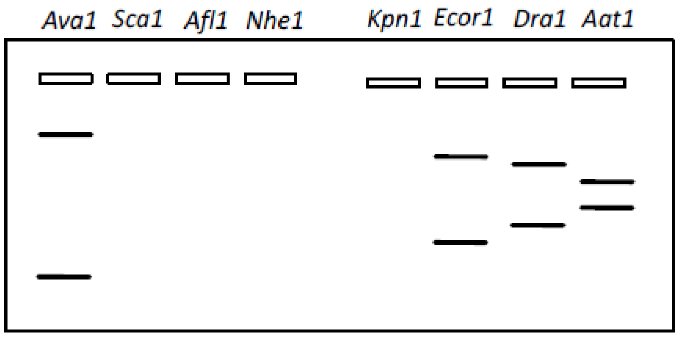

Fig. 1.2

(i) With reference to Fig. 1.1 and 1.2, describe and explain the band pattern of the restriction digest for the recombinant cell. [3]

(ii) Hence identify the site of DNA breakage where crossing over took place in the mutated allele. [1]

Total:[15]

Question 2

For Examiner's use

The herpes simplex virus is an enveloped DNA virus. Viral DNA enters the nucleus of the host cell via nuclear pores and directs synthesis of viral RNA and DNA. The virus is able to grow in non-dividing cells because its genome encodes enzymes involved in synthesising dexoyribonucleotides required for DNA replication.

(a) PCR is used to detect herpes simplex infections in tissue samples. A pair of primers is used to specifically amplify a 148 bp region of the herpes simplex virus genome.

Explain how this allows for the detection of herpes simplex infections in tissue samples. [2]

(b) Describe **two** limitation of PCR. [2]

(c) Describe **one** advantage of PCR. [1]

Total:[5]

For Examiner's use

Question 3

In 2014, a team of scientists claimed that they were able to reprogramme somatic cells into pluripotent stem cells through exposure to stimuli. These resultant stem cells were termed "stimulus-triggered acquisition of pluripotency (STAP) stem cells".

Fig. 3.1 shows how a study was conducted to test if STAP stem cells (STAP SCs) are pluripotent. Somatic cells were isolated from mice and exposed to an acidic environment to form STAP SCs. The STAP SCs were then injected into an early-stage mouse embryo.

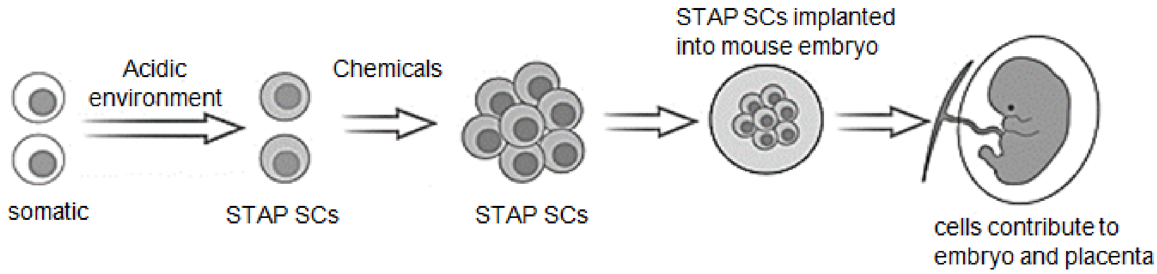

Fig. 3.1

With reference to **Fig. 3.1**,

(a) (i) state the role of the acidic environment in the formation of STAP SCs. [1]

....

....

(ii) describe the unique features of STAP SCs. [2]

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....

(b) Explain whether this study shows that the STAP SCs are pluripotent. [2]

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....

For Examiner's use

(c) Tissue culture of plants involve similar principles to STAP and it has many advantages such as extremely high multiplication rates and being able to be carried out throughout the year independent of seasons.

Describe how tissue culture can be carried out to form a whole new plant. [5]

Total: [10]

For Examiner's use

Question 4

(a) YieldGard, a type of Bt corn that produces a single type of Bt protein, was planted in Midsouth, USA. A Bt resistance monitoring programme showed that the frequency of resistance to the Bt protein in YieldGard increased significantly in sugarcane borers (SCBs). SCBs not only feed on sugarcane, but also other crops such as corn.

To counter the Bt-resistant SCBs, Bt corn expressing multiple types of Bt proteins were recently employed in a greenhouse trial. The trial included a non-Bt control corn and three types of Bt corn:

•Non-Bt control corn – does not produce Bt protein,
•YieldGard®– produces a single type of Bt protein,
•Genuity® VT Triple Pro™ – produces three types of Bt proteins, and
•Genuity® SmartStax™ – produces six types of Bt proteins.

Newly hatched larvae of three SCB populations were manually placed on mature corn plants in the greenhouse. The larvae often bore into the corn stalk and develop tunnels within them. Larval tunnelling by borers inside each stalk was recorded after 21 days as shown in **Fig. 4.1**.

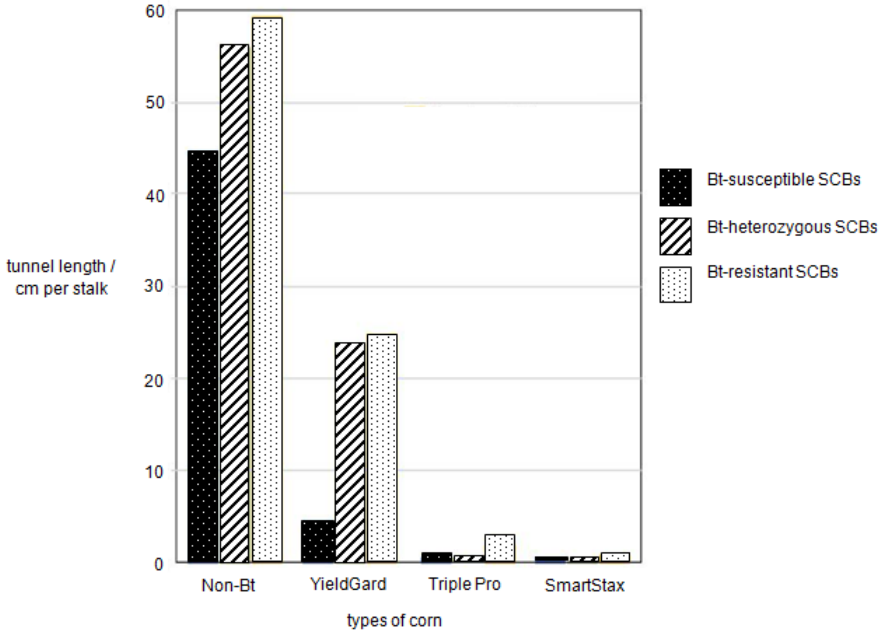

Fig. 4.1

With reference to **Fig 4.1,**

(i) Suggest and support with data which type of Bt corn will best increase crop yield. [2]

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For Examiner's use

(ii) Explain why YieldGard is no longer optimal for cultivation when all three types of SCBs are present. [2]

(iii) Discuss the environmental, ethical or social implications of the above genetically modified Bt plants. [3]

(b) Discuss the need to compare among the three different populations of SCBs. [2]

(c) Pesticide resistance describes the decreased susceptibility of a pest population to a pesticide that was previously effective at controlling the pest. Pest species evolve pesticide resistance via natural selection; the most resistant specimens survive and pass on their genetic traits to their offspring.

Suggest how Triple Pro and SmartStax effectively manages Bt resistance in SCBs. [1]

Total: [10]

Section B: SPA Planning Task (12 marks)
Write your answers on the writing paper provided.
A **NIL RETURN** is required.

Question 5

Yeast undergoes respiration, breaking down glucose into carbon dioxide and water. This process is catalysed by enzymes. Methylene blue is an artificial hydrogen acceptor which is blue in the oxidised form and colourless when reduced.

coloured methylene blue $\xrightarrow{\text{reduction}}$ colourless methylene blue

You are required to plan, but **not** carry out, an investigation on the effect of temperature on the rate of yeast respiration.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- yeast suspension
- glucose solution
- methylene blue

- boiling tubes
- stopwatch
- thermostatically controlled water bath
- beakers
- a variety of different sized measuring cylinders, syringes and pipettes for measuring volumes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible;
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

Total: [12]

Section C: Free-Response Question (20 marks)

Write your answers on the writing paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

Question 6

(a) With reference to genetic diseases, describe cystic fibrosis. [7]

(b) Describe the treatment of severe combined immunodeficiency (SCID), using the viral gene delivery system. [7]

(c) Describe the factors that keep gene therapy from becoming an effective treatment. [6]

Total: [20]

END OF PAPER

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DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2014
YEAR SIX
H2 BIOLOGY (9648)
PAPER 3

<u>Suggested Answers</u>

<u>Structured Questions Answers</u>

1(a)(i)
To (protect) against foreign DNA by digesting/cutting/hydrolyzing it;

(ii)
Mutation and thus changes in DNA sequences;
Creates / destroy / changes new restriction sites;

(iii)
Transformed cells with recombinant DNA will survive in the presence of neomycin whilst transformed cell without recombination will not survive;
Neo gene from vector would come under the control of an enhancer;
High expression of neo gene confers resistance against neomycin;.
No /low expression of neo gene due to absence of enhancer in linearized plasmid / absence of Neo gene on genomic fragment;

(iv)
Cut DNA fragment with restriction enzyme *Hpa1;*
separate DNA fragment via gel electrophoresis;
Ref to southern blotting / transfer to nitrocellulose membrane;
Add radioactive probe complementary to enhancer sequence follow by autoradiography;
A band close to well will represent genomic fragment in non-transformed cell, no band observe for transformed cells with plasmid, and a band further away from well will represent recombinant DNA in transformed cells;

4max

(b)
(i)
presence of 2 restriction fragments each using Ava1, Ecor1, Dra1 and Aat1 digest, indicates presence of restriction sites on recombinant DNA;

digestion with Sca1, Afl1, Nhe1 and Kpn1 yields no bands, indicates absence of restriction sites on recombinant DNA;

Ref. to explanation linking action of named restriction enzyme/position of a named restriction site (Ava1, Ecor1, Dra1 and Aat1) on the size of the fragments produced;

(ii)
Between Ava1 and Sca1;

2
(a)
Primers will only bind to complementary sequences which are unique to the herpes simplex virus;
Ref. hence leading to production of the 148 bp fragment;
Ref. to absence of herpes simplex virus means that no DNA will be amplified; @ 1m

2max

(b)
Very sensitive, any contamination may be amplified as well;
Taq polymerase does not have proof-reading ability, resulting in mutation in PCR product;
A part of the sequence of target must be known in order to design primers;
Only work for relatively short DNA sequence;

(c)
Fast, 30 cycles in 3hr;
Easy to carry out in thermocycler;
capable of amplifying sequences from minute amounts of DNA;
A broad range of nucleic acid sources are suitable templates for PCR amplification;

3(a)(i)

Provides a strong shock / stimulus that causes the cell to dedifferentate;

3(a)(ii)

STAP stem cells are able to reproduce itself and produce many STAP stem cells;

STAP SCs are undifferentiated cells which are able to differentiate under specific conditions into an embryo which contains all the cells of an organism;

3(b)

No, SCs are totipotent as it can also differentiate to form placenta as well as the embryo;

Pluripotent cells have the ability to differentiate into almost any cell type to form any organ or type of cell and so are not totipotent but are multipotent;

3(c)

choose a part of the parent plant such as shoot tip which contains meristematic cells;

sterile conditions and give example; use laminar hood / sodium hypochlorite;

nutrient medium to encourage dedifferentiation and mitosis to produce callus;

Increase auxin to cytokinin ratio in medium to encourage root differentiation / Increase cytokinin to auxin ratio in medium to encourage shoot differentiation;

Grown in hardening medium/sterile soil for acclimatization;

4(a)(i)

SmartStax;

The tunnel length / cm per stalk by all three types of sugarcane borers (SCB) are the lowest as compared to other corn, very close to zero indicating the least damage done to the crop and hence highest yield;

(a)(ii)

Bt-heterozygous SCBs and Bt-resistant SCBs cause significant damage to YieldGard with 25 cm tunnel length as compared to 5 cm of Bt-susceptible SCBs;

Bt-heterozygous SCBs and Bt-resistant SCBs will increase in population size due to being at selectively advantage in the presence of YieldGard and there will still be significant damage to the corn;

4(a)(iii)

Bt plants may cross pollinate with their wild relatives, spreading the Bt gene and causing genetic pollution.

The hybrid plants that contain the Bt gene may kill off other unintended insects and affect ecological balance;

Using 3 or more Bt toxin in Triple Pro and SmartStax may result in accumulation of high amounts of Bt toxins in SCBs. Predators and parasites of SCBs which are of benefit to agriculture could die due to the high levels of toxin;

Triple Pro and SmartStax contains many new transgenic proteins and when consumed by humans may lead to unexpected allergic reactions;

AVP but must be relevant to Bt corn;

3 max

4(b)

Bt-susceptible SCBs acts as a negative control / Bt-resistant SCBs acts as positive control;

Comparison between Bt-resistant and Bt-susceptible populations allows determination of the effect of BT resistance on tunnel length / cm per stalk by SCBs;

4(c)

Having multiple types of Bt toxin proteins present in one plant ensures all the SCBs are killed off, so that no Bt-resistant SCB survive to pass on the advantageous gene to its offspring;

Planning Answer

Theory	Yeast undergoes respiration in the presence of glucose. NAD^+ and FAD are reduced to form NADH and $FADH_2$ during glycolysis, link reaction and Krebs cycle. The H+ released reduces blue Methylene blue to coloreless. The time taken for decolouration of methylene blue can be used to measure the rate of respiration. As temperature increases, kinetic energy of respiratory enzymes and glucose increase. This Increases frequency of effective collision between respiratory enzymes and glucose molecules, more enzyme-substrate complexes are formed and this rate of reaction increases. As temperature increase, rate of reaction increase, time taken for decolouration of methylene blue will be shorter. Beyond optimum temperature, enzyme will be denatured since weak bond e.g. hydrogen and ionic bonds are broken by heat, active site no longer fit glucose. Rate of reaction decreases, time taken for decolouration of methylene blue will be longer.	✓1 description of scientific reasoning ✓2 Theory of the method used and expected relationship between [penicillin] and diameter of clear zone
Variables	Independent variable: temperature: 5 temperatures of: 20, 30, 40, 50, 60^0C, to be used for experiment using thermostatically controlled water bath. Dependent variables: Time taken for methylene blue to decolourise using stop watch and visual judgemet Controlled variables for all experiments: Volume of glucose, keep constant at 10mL Volume of yeast used in reaction, keep constant at 2mL Volume of methylene blue used, keep constant at 5 drops.	✓ 3 independent, dependent variables ✓ 4 controlled variables
Procedure	1. relevant diagram 2. Use a 10mL syringe to measure out 10mL of glucose solution and put it into a test tube. 3. Use a 5mL syringe to measure out 2mL of yeast suspension and put it in another test tube. Be sure to stir the yeast suspension before using. Use a dropper to add 5 drops of methylene blue into the same test tube. 4. Equilibrate temperature of the glucose and enzyme by putting them into the thermostatically-controlled water bath of 20^oC for 3 minutes. 5. At the end of 3 minutes, add the glucose into the yeast test tube and immediately start the stop watch. Ensure that test tube with mixture is in water bath. 6. Record the time taken for methylene blue to decolorize in the table below.	✓ 5 diagram ✓ 6 volume ✓ 7 Equilibrate temperature ✓ 8 Ensure temperature constant throughout reaction ✓9 record time taken

<table>
<tr>
<td></td>
<td>
<table>
<tr><th rowspan="2">Temperature /^{o}C</th><th colspan="4">Time taken for methylene blue to decolorize / s</th></tr>
<tr><th>Reading 1</th><th>Reading 2</th><th>Reading 3</th><th>Average</th></tr>
<tr><td>20</td><td></td><td></td><td></td><td></td></tr>
<tr><td>30</td><td></td><td></td><td></td><td></td></tr>
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<tr><td>60</td><td></td><td></td><td></td><td></td></tr>
</table>

7. Repeat steps 1 – 5 again to get 2 more readings and calculate the average time taken.
8. Repeat steps 1 – 6 again for the rest of the temperatures.
9. Conduct 2 replicates to ensure consistency and reproducibility.
10. Plot graph of Time taken for methylene blue to decolorize / s against Temperature /^{o}C.
</td>
<td>✓ <u>10</u> appropriate table with correct headings with units for recording time taken

✓ <u>11</u> replicates, 2 repeats

✓ <u>12</u>graph</td>
</tr>
<tr>
<td>Risk and precaution</td>
<td>Thermostatically water bath is an electrical appliance, so use with care to prevent electrocution.</td>
<td>✓ <u>13</u> Risk and precaution</td>
</tr>
</table>

Essay Answers

(a)	With reference to genetic diseases, describe cystic fibrosis. [7]
Marking Point	**Answer**
1	Autosomal recessive disease. Both copies of the cystic fibrosis transmembrane conductance regulator (CFTR) genes are defective for disease phenotype to show;
2	There is a deletion of three nucleotides coding for phenylalanine in the CFTR gene on chromosome 7;
3	The translated CFTR protein is unable to be transported to the membrane or is non-functional in transporting chloride ions out of the epithelial cells of the lungs;
4	Accumulation of chloride ions causes water to enter these cells resulting in thick, sticky mucus that clogs the airways in lungs and ducts of other organs;
5	Bacterial infections can also occur as a result of the mucus and lead to pneumonia and respiratory failure;
6	The patient's pancreatic duct is choked by thick mucus which prevents the release of enzymes, leading to indigestion, and a reduction in the absorption of digested food and thus malnutrition;
7	Males also experience infertility owing to problems related to ion transport in male reproductive organs;

(b)	Describe the treatment of severe combined immunodeficiency (SCID), using the viral gene delivery system. [7]
Marking Point	**Answer**
1	An ex vivo viral mediated gene delivery approach can be used to treat both ADA-SCID and sex-link-SCID;
2	This involves removing diseased T-lymphocytes obtained from the blood of the patient / bone marrow stem cell and stimulated to proliferate in culture;
3	A functional adenosine deaminase (ADA) cDNA / IL2-RG cDNA in a retroviral vector was then used to introduce the normal ADA gene into patient's cells;
4	The retrovirus had been inactivated to become non-pathogenic;
5	The retrovirus integrates the normal ADA gene / IL2-RG gene into the chromosomes of the patient's cells and the gene expression is permanent;
6	The transgenic lymphocytes were allowed to proliferate further in the culture and reintroduced into the patient's body;
7	this step is repeated until the patients has a high enough lymphocyte count to alleviate symptoms;

(c)	Describe the factors that keep gene therapy from becoming an effective treatment. [6]
Marking Point	**Answer**
1	Ectopic insertion of transgene by retrovirus may cause displacement of other genes and cause gene disruption harming the patient;
2	Insertional mutagenesis by retroviruses may cause tumour formation due to disruption of a tumour suppressor gene or over expression of proto-oncogenes;
3	Immune response by the host may attack the virus meant to deliver the cure. An

	enhanced response to invaders it has seen before makes it difficult for gene therapy to be repeated in patients;
4	Virus as vectors may regain virulence. Viruses have shown to be able to recover its ability to cause disease when introduced into patients;
5	Multigene disorders such as high blood pressure, Alzheimer's disease and diabetes are caused by the combined effects of variations in many genes. These disorders would be especially difficult to treat effectively using gene therapy;
6	Inappropriate regulation of gene. In thalassaemia, an over-expression of the transferred gene would cause a new imbalance of globin chains, whereas low levels of gene expression would be ineffective therapeutically.
7	Short-lived nature of gene therapy. For cystic fibrosis, the epithelial cells are replaced by the body every few months. Any therapy would have to be repeated every time the cells are replaced;

6 max